

KIRAN BRAHMADEWARI

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EXPERIENCE

Software Engineer Intern

May 2021 – August 2021; June 2022 – September 2022;

Oracle Cloud Infrastructure (OCI)

May 2023 – August 2023; May 2024 – August 2024

- Built **Java** workflow to automate **Top of Rack (ToR) switch replacement**, reducing **Jira** triage effort by **80%**
- Developed a **Python** command line tool to automate release workflows by generating deployment artifacts, rollback plans, and **Jira** change-management records across multiple environments and cloud realms
- Created a command line tool in **Python** to automatically convert **Terraform** alarm configurations into **Grafana** dashboards, eliminating manual dashboard creation for network and infrastructure health monitoring
- Designed and implemented a **Java** data access layer (DAO) and **Oracle Database** schema to manage fault-code telemetry for downstream data warehouse ingestion
- Onboarded throttling services with **OCI** limits infrastructure using **Terraform** and internal automation tooling
- Developed **automated unit tests** achieving **90% method coverage** and authored technical documentation (one-pagers, design docs, etc.) in **Confluence**
- Teams: **Virtual Network Interface Controller as a Service, Asset Repair Orchestration, Block Storage Management Plane**

Research Assistant

September 2023 – April 2024

Analytics for Cyber Defense Lab

- Developed a **Python** client-server system for privacy-preserving distributed large matrix multiplication across **Internet of Things (IoT)** edge devices, using matrix partitioning, parallel execution, and homomorphic encryption-based Multi-Party Computation (MPC); faster runtime than **NumPy** for sufficiently large matrices

PROJECTS

AI Detection System for Real-Time Environmental Monitoring

January 2024 – May 2024

Capstone II

- Implemented an **HTTP MJPEG streaming server** in **C++** and **C** for low-latency video ingestion from **IoT** devices, enabling real-time data flow for edge inference pipelines
- Architected a **Python**-based edge computer vision pipeline using **PyTorch**, **YOLOv8**, and **OpenCV** to process live video streams from **ESP32-CAM** and smartphones for harmful algae detection in natural water systems
- Automated a machine learning pipeline in **Google Colab** using **Roboflow** and **YOLOv8**, including dataset preprocessing, augmentation, and model training, achieving **89% mAP**, **82.6% precision**, **89.4% recall**
- Led a 3-person engineering team as **Tech Lead**, defining system architecture, implementation plan, and **Software Development Life Cycle (SDLC)** workflow for an end-to-end AI-based environmental monitoring system

Windows-macOS Compatibility Layer

March 2025 – Present

blog.lynkos.dev/posts/play-windows-games

- Built a **Bash**-based configuration toolkit and application launcher for running Windows games and software on **Apple Silicon**, integrating **WINE** and **Game Porting Toolkit (GPTk)** with **DXMT**, **DXVK**, and **MoltenVK** to simplify compatibility-layer execution workflows

Open Source Contributor — Chirpy Jekyll Theme

January 2026 – April 2026

PR #2642: fix: sidebar email link

- Fixed a Firefox-specific navigation bug in the open-source **Jekyll** theme jekyll-theme-chirpy (**10.1k+ GitHub stars**), resolving a sidebar redirect issue and contributing a merged pull request to a widely used static site template

SKILLS

Languages: Python, Java, JavaScript, TypeScript, Bash, SQL, C, C++, HTML, CSS, SASS, Ruby

Frameworks & Libraries: Spring, Node.js, React, JUnit, Mockito, OpenCV, PyTorch, TensorFlow, Keras

Tools: Git, Linux, Docker, GitHub Actions, Terraform, Jira, Maven, Grafana, Prometheus, MongoDB, Firebase

EDUCATION

Bachelor of Science in Computer Science

Graduated August 2025

Florida International University